Wrangling Results

in this project i assessed the data by two ways programmatically and visually and i extracted quality and tidiness issues as possible and here is the results of my assessing

Twitter dataframe

Data quality issues extracted from the dataset:

1- there was a lot of unneeded columns that will not add any value to the analysis or increase its accuracy or even the modeling

2- in a row in the denominator_rating column the denominator was zero and this isn't logical

3- there was missing values (nan values) in columns

4- timestamp column was not in datetime data type

5- there was a name in the name column that was likely written incorrectly and that name is Juckson and it was expected to be Jackson

6- source column which has values like the source where the tweet was published from was in <a> HTML tag format

7- in the dataset there were retweets that was needed to be removed from the dataset to insure accuracy and the efficiency of the analysis

Tidiness issues extracted from the dataframe:

1- text column has the link of the tweet and this violates the tidiness rule of the dataframe that each variable is a column and this column can be used for merging along with id column

2- columns like doggo,floofer and others can be grouped together in one column for analysis

3- from the source column i can extract the source of each tweet

Tweet dataframe

Data quality issues extracted from the dataset:

1- id_str is in string data type

2- the presence of missing values

3- present of retweets

Tidiness issues extracted from the dataframe:

1- display text range column is in two variables the display text range min and max

2- urls can be extracted from the text column for merging

How was these issues fixed

first lets start with the twitter dataframe:

first issue was the retweets i fixed it by choosing only the rows with nan values in the retweets column as they mean that there was no retweets in this row

then i merged all the columns of dog types into one column by coding a function that can do this

then i coded a function that extracts all the columns that has missing values more than 50% then i dropped them to insure a clean dataset and to easily extract information from the dataset

then the name column that had missing values i fiiled it with Unknown then in the name column there was some values that are not related to the column i replaced them with Unknown and fixed the issue in the Juckson name

then i converted the data type of the timestamp column to dateTime data type

then from the source column i extracted the source of the tweet and from the text column i extracted the link of the tweet and made columns for them and replaced the zero in the rating_denominator with the mean rounded to the nearest value

now with the tweet dataframe:

i got rid of the retweets by the same approach i did in the twitter dataframe then i used the same function to extract the nan columns with 50% nan values and drop them then i replaced the nan values with none in the object data type columns and got rid of the nan values in other columns by putting the mean then i converted the id_str column to string data type and extracted the minimum and maximum using the display_text_range column then i extracted the link of the tweet by the text column